

Assignment 1 Solutions

NIS3352-01 Modern Cryptography (2)

Problem 1. Let $F_1, \dots, F_p$ be events, where $p = \text{poly}(\lambda)$, and suppose that for every $i \in [p]$,

$$\Pr[F_i] = 1 - \text{negl}_i(\lambda)$$

for some negligible function negl_i . We show that

$$\Pr[F_1 \wedge F_2 \wedge \dots \wedge F_p]$$

is overwhelming.

Let $\overline{F_i}$ denote the complement of F_i . Then

$$\Pr[\overline{F_i}] = 1 - \Pr[F_i] = \text{negl}_i(\lambda).$$

By De Morgan's law,

$$\Pr[F_1 \wedge \dots \wedge F_p] = 1 - \Pr[\overline{F_1} \vee \dots \vee \overline{F_p}].$$

Applying the union bound,

$$\Pr[\overline{F_1} \vee \dots \vee \overline{F_p}] \leq \sum_{i=1}^p \Pr[\overline{F_i}] = \sum_{i=1}^p \text{negl}_i(\lambda).$$

Since $p = \text{poly}(\lambda)$, the sum of polynomially many negligible functions is still negligible. Hence there exists a negligible function $\text{negl}'(\lambda)$ such that

$$\Pr[\overline{F_1} \vee \dots \vee \overline{F_p}] \leq \text{negl}'(\lambda).$$

Therefore,

$$\Pr[F_1 \wedge \dots \wedge F_p] \geq 1 - \text{negl}'(\lambda),$$

which means that $\Pr[F_1 \wedge \dots \wedge F_p]$ is overwhelming.

Problem 2. We prove that IND-CPA security implies IND-mCPA security.

Let A be any PPT adversary in the IND-mCPA experiment. Suppose A outputs two message vectors

$$M_0 = (m_{0,1}, \dots, m_{0,t}), \quad M_1 = (m_{1,1}, \dots, m_{1,t}),$$

where $t = t(\lambda)$ is polynomially bounded.

For each $j \in \{0, 1, \dots, t\}$, define the hybrid experiment H_j as the experiment in which the challenger returns encryptions of

$$(m_{1,1}, \dots, m_{1,j}, m_{0,j+1}, \dots, m_{0,t}).$$

Thus:

$$H_0 \text{ encrypts } M_0, \quad H_t \text{ encrypts } M_1.$$

Let

$$p_j := \Pr[A \text{ outputs } 1 \text{ in } H_j].$$

We first prove the following claim.

Claim. For every $j \in \{0, \dots, t-1\}$, the distributions H_j and H_{j+1} are computationally indistinguishable. Equivalently,

$$|p_{j+1} - p_j|$$

is negligible.

Proof of the claim. Fix any $j \in \{0, \dots, t-1\}$. We show that if there exists a PPT distinguisher between H_j and H_{j+1} with non-negligible advantage, then one can construct a PPT adversary B that breaks IND-CPA security.

Adversary B receives a public key $\mathbf{pk}$ from the IND-CPA challenger and gives it to A . Adversary A outputs two message vectors

$$M_0 = (m_{0,1}, \dots, m_{0,t}), \quad M_1 = (m_{1,1}, \dots, m_{1,t}).$$

Then B submits the pair

$$(m_{0,j+1}, m_{1,j+1})$$

to its IND-CPA challenger and receives a challenge ciphertext C^* , which is an encryption of one of these two messages.

Now B constructs a ciphertext vector for A as follows:

- for positions $1, \dots, j$, it honestly encrypts $m_{1,1}, \dots, m_{1,j}$;
- for position $j+1$, it uses the challenge ciphertext C^* ;
- for positions $j+2, \dots, t$, it honestly encrypts $m_{0,j+2}, \dots, m_{0,t}$.

If C^* is an encryption of $m_{0,j+1}$, then the resulting distribution is exactly H_j . If C^* is an encryption of $m_{1,j+1}$, then the resulting distribution is exactly H_{j+1} .

Therefore, any non-negligible distinguisher between H_j and H_{j+1} immediately yields a non-negligible IND-CPA adversary B , contradicting the IND-CPA security of the encryption scheme. Hence H_j and H_{j+1} are computationally indistinguishable, proving the claim.

By the claim, for every $j \in \{0, \dots, t-1\}$, the quantity

$$|p_{j+1} - p_j|$$

is negligible. Since

$$p_t - p_0 = \sum_{j=0}^{t-1} (p_{j+1} - p_j),$$

the triangle inequality gives

$$|p_t - p_0| \leq \sum_{j=0}^{t-1} |p_{j+1} - p_j|.$$

Because t is polynomially bounded and each term on the right-hand side is negligible, the entire sum is negligible. Hence

$$|p_t - p_0|$$

is negligible.

But p_0 is exactly the probability that A outputs 1 when given encryptions of M_0 , and p_t is exactly the probability that A outputs 1 when given encryptions of M_1 . Therefore A 's IND-mCPA advantage is negligible. This proves that the encryption scheme is IND-mCPA secure.

The converse direction is immediate, since IND-CPA is simply the special case of IND-mCPA in which the message vectors have length 1. Therefore, IND-CPA and IND-mCPA are equivalent.

Problem 3. We show that the plain RSA encryption scheme is not IND-CPA secure.

The key observation is that plain RSA encryption is deterministic: for a fixed public key (N, e) and a message $M \in \mathbb{Z}_N^*$, the ciphertext is always

$$C = M^e \bmod N.$$

There is no randomness in the encryption algorithm.

We now describe an adversary $\mathcal{A}$ that wins the IND-CPA game with probability 1.

1. The challenger generates an RSA key pair and gives the public key (N, e) to $\mathcal{A}$.
2. $\mathcal{A}$ chooses two distinct messages $M_0, M_1 \in \mathbb{Z}_N^*$ and sends them to the challenger.
3. The challenger chooses a random bit $b \in \{0, 1\}$, computes

$$C^* = M_b^e \bmod N,$$

and sends C^* to $\mathcal{A}$.

4. Using the public key, $\mathcal{A}$ computes

$$C_0 = M_0^e \bmod N.$$

If $C^* = C_0$, then $\mathcal{A}$ outputs 0; otherwise it outputs 1.

Because RSA encryption is deterministic and $M_0 \neq M_1$, the ciphertexts of M_0 and M_1 are distinct. Hence $\mathcal{A}$ always recovers the correct bit b . Its success probability is therefore 1, so its advantage is also maximal.

Thus plain RSA encryption is not IND-CPA secure.

Problem 4. Consider the encryption scheme with public key (G, q, g, h) and private key x , where $h = g^x$. To encrypt a bit $b \in \{0, 1\}$:

- if $b = 0$, choose $y \xleftarrow{\$} \mathbb{Z}_q$ and output $(c_1, c_2) = (g^y, h^y)$;
- if $b = 1$, choose independent $y, z \xleftarrow{\$} \mathbb{Z}_q$ and output $(c_1, c_2) = (g^y, g^z)$.

Efficient decryption. Given the private key x and a ciphertext (c_1, c_2) , compute c_1^x and compare it with c_2 .

Indeed:

- If $b = 0$, then

$$c_1^x = (g^y)^x = g^{xy} = (g^x)^y = h^y = c_2.$$

So the equality always holds.

- If $b = 1$, then $c_2 = g^z$ for an independent random z , while

$$c_1^x = (g^y)^x = g^{xy}.$$

Since z is independent and uniform in $\mathbb{Z}_q$, the probability that $g^z = g^{xy}$ is exactly $1/q$, which is negligible in the security parameter.

Therefore the decryption algorithm is:

output 0 if $c_2 = c_1^x$, and output 1 otherwise.

This is efficient and correct except with negligible error probability.

CPA security under the DDH assumption. We prove that if DDH is hard in G , then this encryption scheme is IND-CPA secure.

Suppose, toward contradiction, that there exists a PPT adversary $\mathcal{A}$ that distinguishes encryptions of 0 and 1 with non-negligible advantage. We build a PPT distinguisher $\mathcal{B}$ for DDH.

The DDH challenger gives $\mathcal{B}$ a tuple

$$(g, A, B, T) = (g, g^x, g^y, T),$$

where either $T = g^{xy}$ (the Diffie–Hellman case) or $T = g^z$ for uniform $z \xleftarrow{\$} \mathbb{Z}_q$ (the random case).

Algorithm $\mathcal{B}$ proceeds as follows:

1. Set $h := A = g^x$ and give the public key (G, q, g, h) to $\mathcal{A}$.
2. Give $\mathcal{A}$ the challenge ciphertext

$$(c_1, c_2) := (B, T).$$
3. Let b' be the bit output by $\mathcal{A}$. Output “DDH” if $b' = 0$, and output “random” if $b' = 1$.

Now analyze the simulation:

- If $T = g^{xy}$, then

$$(c_1, c_2) = (g^y, g^{xy}) = (g^y, h^y),$$

which is distributed exactly as an encryption of 0.

- If $T = g^z$ for uniform z , then

$$(c_1, c_2) = (g^y, g^z),$$

which is distributed exactly as an encryption of 1.

Thus any non-negligible advantage of $\mathcal{A}$ in distinguishing encryptions of 0 and 1 translates directly into a non-negligible advantage for $\mathcal{B}$ in solving DDH, contradicting the DDH assumption.

Therefore the scheme is IND-CPA secure under the DDH assumption.